

dispatch: task chosen by the selected intensity class or rule, or rest

Simulated workday (POMDP)

Workload generator

arrivals, difficulty d_i ,
work w_i , deadline D_i

Latent cognitive state

homeostatic attention H_t
fatigue F_t , circadian $c(h_t)$

Productivity model

$$\rho_t = \rho_0 A_t^{0.5+d} (1 - \kappa F_t)$$

Multi-objective reward

$$r_t = r_t^{\text{task}} + \mu A_{t+1} - \lambda F_{t+1}$$

sensing

Noisy observation o_t

$$\tilde{A}_t = A_t + \epsilon_t^A, \tilde{F}_t = F_t + \epsilon_t^F$$

time + per-class queue summary

History window

$$x_t = [o_{t-k+1}, \dots, o_t]$$

Attention-aware DQN agent

ϵ -greedy action

$$a_t \in \{\text{easy, hard, EDF, SPT, break}\}$$

Dueling Q-network

$$Q(x_t, \cdot; \theta) = V(x_t) + A(x_t, \cdot)$$

2×128 ReLU

Replay buffer $\mathcal{D}$

$$(x_t, a_t, r_t, x_{t+1})$$

Double-DQN target

$$y = r + \gamma Q_{\bar{\theta}}(x', \arg \max_{a'} Q_{\theta}(x', a'))$$

Huber TD
loss, Adam